

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
Given : Mass of the substance (m) = 1 liter = 1000 grams , the mass of aluminum = 1 kg = 1000 grams .
The density of water is 1 g/ml so Specific heat of water (s) = 1 cal/gÂ°C .
Given the specific heat of aluminum (sa) = 215 cal/lb / â€¢ 760 grams (cal) . We also know its latent heat of fusion lat heat_m of steel (dg) = 0 w neustaths .
</think>< answer> A . 28 Kcal , 6 . 3 Kcal </answer>

